UPI Architectural Diagram & Specification

Technical System End-to-End Core Workflow Document

1. Unified Payments Interface (UPI) ASCII Architecture Diagram

The following structural map models the real-time request-response message processing loop among all structural components of the open-loop ecosystem:

```
+------------------+                   +------------------+
|                  |     (1) Scan      |                  |
|    CUSTOMER      |------------------>|     MERCHANT     |
|                  |                   |                  |
+------------------+                   +------------------+
       |        ^                                ^
       |        |                                |
   (2) |    (8) | Notification               (8) | Notification
 Input |  Success                                | via Callback
  PIN  |  Callback                               |
       v        |                                |
+------------------+                   +------------------+
|   GOOGLE PAY     |                   |                  |
|     (TPAP)       |                   | ACQUIRER BANK    |
+------------------+                   | (Beneficiary Bank|
       |        ^                      |     or PSP)      |
   (3) |    (8) |                      +------------------+
ReqPay |  RespPay                                ^
       v        |                                |
+------------------+                             | (7) Credit
| REMITTER BANK    |                             |   Account
|   (Sponsor /     |                             |
|  Remitter PSP)   |                             |
+------------------+                             |
       |        ^                                |
   (4) |    (8) |                                |
ReqPay |  RespPay                                |
       v        |                                |
+----------------------------------------------------------+
|                  NPCI CENTRAL SWITCH                     |
|       [ Central VPA Mapper | Routing Infrastructure ]    |
+----------------------------------------------------------+
       |        ^
   (5) |    (6) | Debit
ReqTxn |  Success
   Pin |  RespPay
       v        |
+------------------+
|   ISSUER BANK    |
| (Remitter CBS &  |
|    HSM Engine)   |
+------------------+
```

2. Architectural Component Specification

Customer: The end consumer initiating the real-time asset transfer. They access the platform network nodes via an authorized client device, provide hardware intent mapping via biometrics, and validate secure payloads utilizing their encrypted UPI PIN.

Google Pay (TPAP): A Third-Party Application Provider running on the client device. It manages frontend onboarding token flows, validates SIM connectivity benchmarks, strips application containers from raw secure credentials via an isolated client runtime library, and coordinates transaction metadata formatting.

NPCI Central Switch: The core transactional overlay backbone processor. It houses the ultra-low-latency distributed key-value Central Mapper that matches Virtual Payment Addresses (VPAs) directly to structural routing indicators (IFSC, Account Numbers). It validates packet signatures and serves as a strict central broker.

Issuer Bank (Remitter Bank): The financial node housing the origin liquidity. This infrastructure runs an internal Hardware Security Module (HSM) cluster to decrypt the PIN securely and interfaces directly with its Core Banking System (CBS) to perform balance evaluations and atomic account debits.

Acquirer Bank (Beneficiary Bank): The ledger processing hub hosting the recipient merchant's profile directory. This entity parses inbound transfer requests routed from the NPCI switch and performs asynchronous accounting entries before triggering low-latency webhooks.

Merchant: The business entity receiving the fund execution. They provide transaction endpoints (Static/Dynamic QRs or integrated Android intents) and listen for structural payment updates to fulfill orders.

3. End-to-End System Workflow Description

Every transaction sequence moves through the network using secure mTLS connections over specific API schemas:

1. **Step 1: Scan & Payload Parse**

   The Customer opens Google Pay to scan the Merchant's visual layout string. The TPAP app decodes the payload parameters (VPA, Merchant Category Code, Name, Base Amount, and Unique Reference ID) directly from the QR pattern array.

2. **Step 2: Credential Entry**

   The Customer defines the payment value (if using a static QR) and triggers authorization. The embedded, isolated NPCI Common Library SDK intercepts control, takes the user's secret PIN, and encrypts it using the Issuer's asymmetric public key within protected execution memory.

3 Step 3: App-to-PSP Handoff

Google Pay transfers the payment transaction block (`ReqPay` API packet) to its designated Sponsor/Remitter PSP Bank API endpoints over a secured network layer.

4 Step 4: PSP-to-Switch Routing

The Remitter PSP Bank signs the structured message envelope with its institutional security key and routes the transaction directly into the central NPCI processing nodes.

5 Step 5: Address Resolution & Debit Request

NPCI acts as the central router. It parses the target VPA, queries its distributed Central Mapper database to identify the destination bank, and forwards a debit instruction packet (`ReqTxnPin`) to the Issuer Bank's gateway network.

6 Step 6: PIN Validation & Atomic Account Debit

The Issuer Bank redirects the encrypted PIN string to its secure Hardware Security Module (HSM). If the PIN matches and the Core Banking System (CBS) balance validation succeeds, the ledger account is debited immediately. The Issuer then returns a signed success status string (`RespPay`) to NPCI.

7 Step 7: Credit Allocation

Upon recording the successful debit packet, NPCI maps the transactional state to a credit flow and fires a `ReqCredit` API instruction to the Acquirer Bank's internal network. The Acquirer Bank updates the target merchant's account ledger in real time.

8 Step 8: Final Status Broadcast

NPCI broadcasts the completed resolution to both ends of the pipeline simultaneously: the Remitter PSP updates the Google Pay client screen, and the Acquirer Bank fires a callback to update the Merchant's POS soundbox or terminal display.